



School of Computer Science and Engineering

Course Title: Software Engineering Lab

Course Code: BCSE301P

Lab Assessment II

Slot:L3-L4

Class Id: VL2025260504501

Due date:19 Jan 2026

Ex. No: 3 - Requirement Modeling using Entity Relationship Diagram (ERD)

Aim: To identify the key entities, attributes, and relationships for the "MediConnect" system and design an Entity Relationship Diagram (ERD) representing the structural data requirements.

Problem Statement: "MediConnect" Telemedicine Platform The system requires a robust database to manage patients, certified doctors, appointment schedules, video consultation sessions, digital prescriptions, and secure payment transactions. The data model must enforce strict relationships (e.g., a prescription cannot exist without an appointment) and handle sensitive medical attributes.

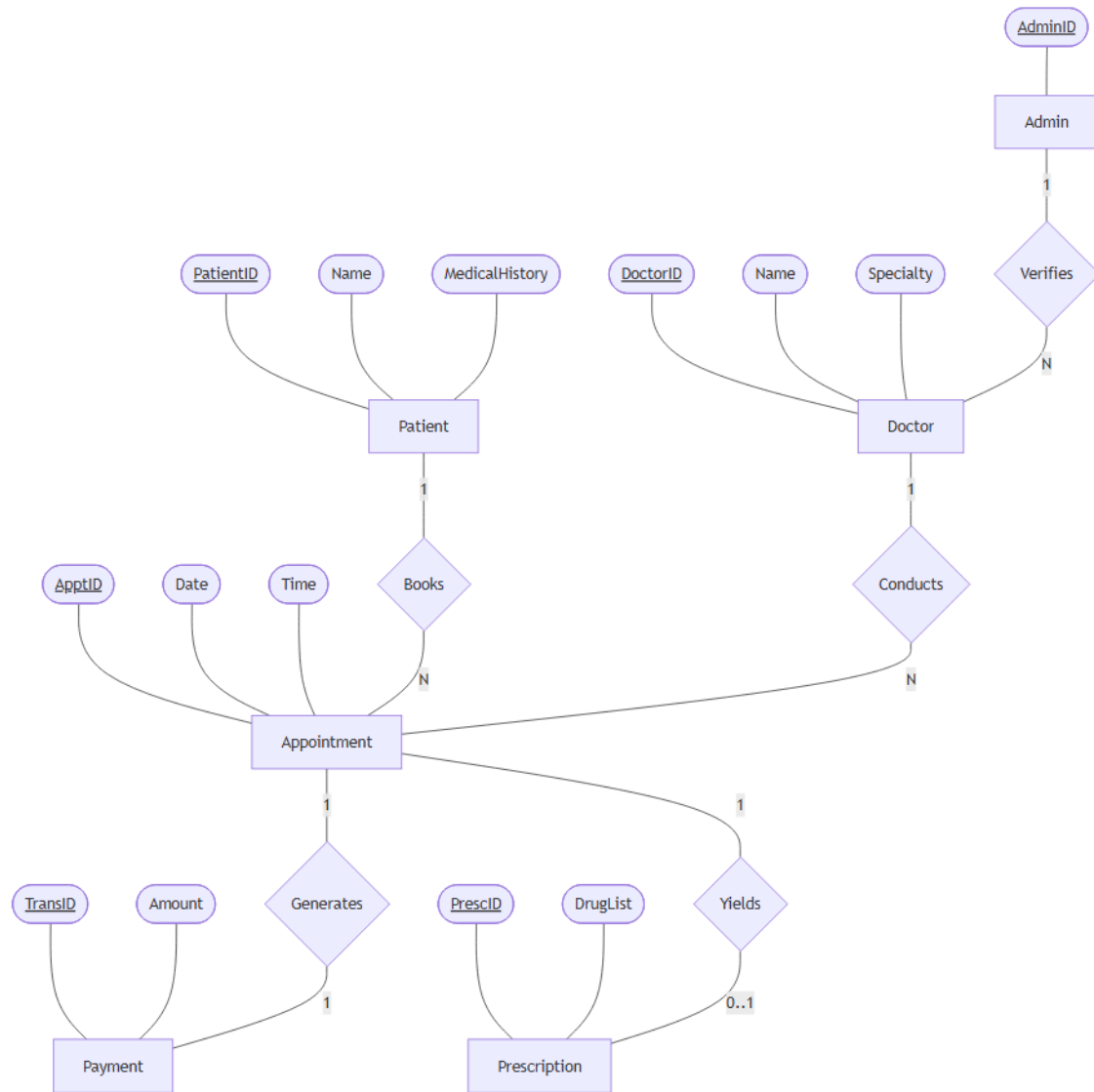
1. Identification of Entities and Attributes:

- **Patient:** PatientID (PK), Name, MedicalHistory.
- **Doctor:** DoctorID (PK), Name, Specialty.
- **Appointment:** ApptID (PK), PatientID (FK), DoctorID (FK), Date, Time.
- **Prescription:** PrescriptionID (PK), ApptID (FK), DrugList.
- **Payment:** TransactionID (PK), ApptID (FK), Amount.
- **Admin:** AdminID (PK), Name, Role.

2. Identification of Relationships:

- **Books:** A **Patient** books one or many **Appointments** (1:N).
- **Conducts:** A **Doctor** conducts one or many **Appointments** (1:N).
- **Generates:** An **Appointment** generates exactly one **Payment** (1:1).
- **Issues:** An **Appointment** results in zero or one **Prescription** (1:0..1).
- **Verifies:** An **Admin** verifies many **Doctors** (1:N).

3. Entity Relationship Diagram (ERD):



Ex. No: 4 - REQUIREMENT MODELING USING DFD (FUNCTIONAL MODELING)

AIM: TO MODEL THE FUNCTIONAL FLOW OF DATA WITHIN THE "MEDIConnect" SYSTEM USING CONTEXT FLOW DIAGRAMS AND DATA FLOW DIAGRAMS (LEVEL-0 AND LEVEL-1).

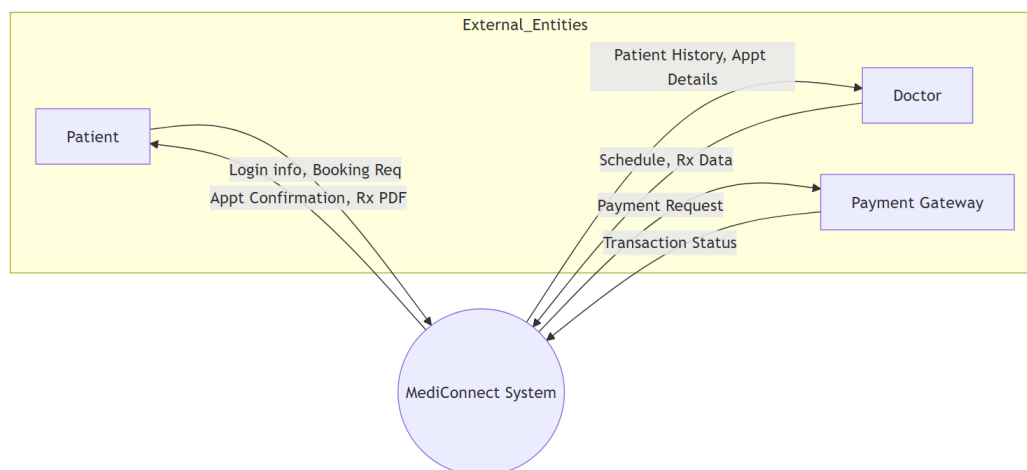
A) CONTEXT FLOW DIAGRAM (LEVEL 0)

REPRESENTS THE ENTIRE "MEDIConnect" SYSTEM AS A SINGLE PROCESS INTERACTING WITH EXTERNAL ENTITIES.

EXTERNAL ENTITIES:

- **PATIENT:** SENDS REGISTRATION DATA, BOOKING REQUESTS, AND PAYMENT DETAILS.
- **DOCTOR:** SENDS SCHEDULE AVAILABILITY AND PRESCRIPTION DATA.
- **ADMIN:** SENDS VERIFICATION APPROVALS.
- **PAYMENT GATEWAY (EXTERNAL):** SENDS TRANSACTION STATUS.
- **VIDEO API (EXTERNAL):** SENDS CONNECTION TOKENS.

DIAGRAM:



B) LEVEL-0 DFD

REPRESENTS THE MAJOR INTERNAL PROCESSES AND DATA STORES.

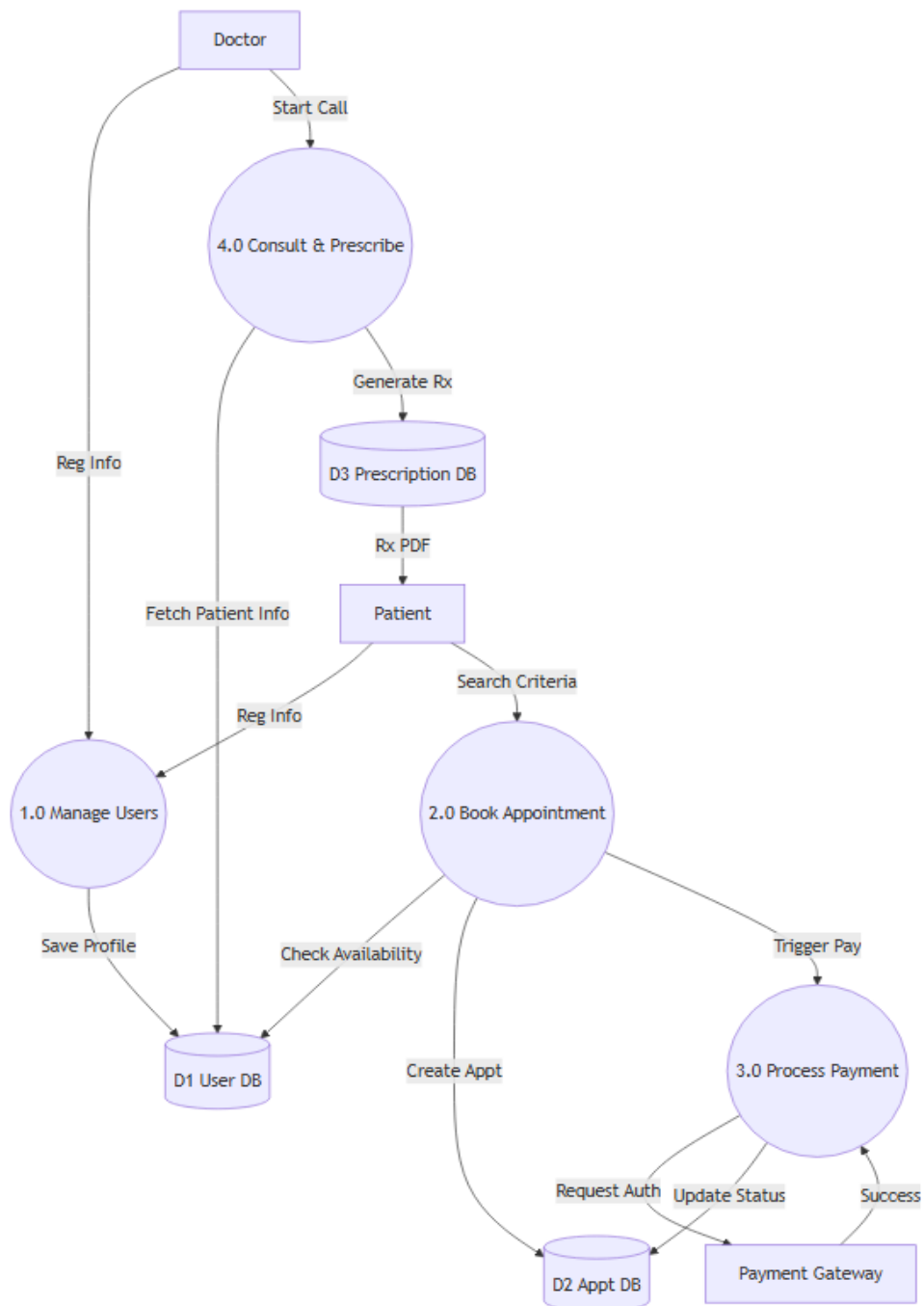
MAJOR PROCESSES:

- **1.0 MANAGE USERS:** HANDLES REGISTRATION AND AUTHENTICATION.
- **2.0 BOOK APPOINTMENT:** HANDLES SEARCH AND SCHEDULING.
- **3.0 PROCESS PAYMENT:** HANDLES FINANCIAL TRANSACTIONS.
- **4.0 CONSULT & PRESCRIBE:** HANDLES VIDEO SESSIONS AND RX GENERATION.

DATA STORES:

- **D1: USER DATABASE**
- **D2: APPOINTMENT DATABASE**
- **D3: PRESCRIPTION DATABASE**

DIAGRAM:



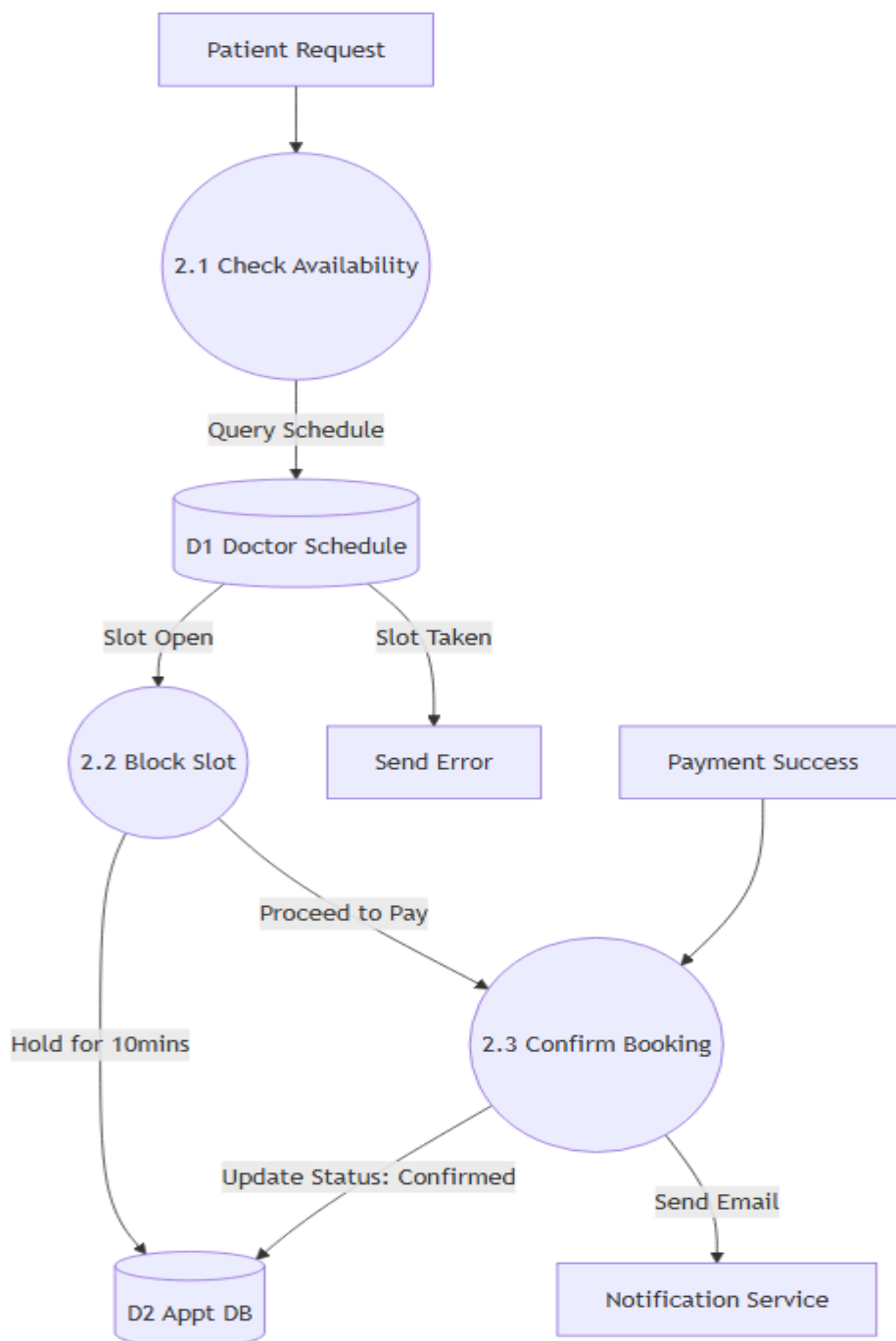
c) LEVEL-1 DFD (DECOMPOSITION OF PROCESS 2.0)

DECOMPOSING "2.0 BOOK APPOINTMENT" INTO GRANULAR SUB-PROCESSES.

SUB-PROCESSES:

- **2.1 CHECK DOCTOR AVAILABILITY:** VERIFIES IF THE SELECTED SLOT IS FREE.
- **2.2 BLOCK SLOT:** TEMPORARILY HOLDS THE SLOT DURING PAYMENT.
- **2.3 CONFIRM BOOKING:** FINALIZES THE RECORD AFTER PAYMENT.

DIAGRAM:



Ex. No: 5 - REQUIREMENT MODELING USING STATE TRANSITION DIAGRAM (STD)

AIM: TO IDENTIFY THE DIFFERENT STATES OF THE SYSTEM AND DRAW A STATE TRANSITION DIAGRAM (STD) SHOWING STATES, EVENTS, AND TRANSITIONS.

CONTEXT: WE WILL MODEL THE **LIFECYCLE OF AN APPOINTMENT** OBJECT, AS IT IS THE CENTRAL DYNAMIC ENTITY IN THE SYSTEM.

1. IDENTIFICATION OF ELEMENTS:

- **INITIAL STATE:** NULL (BEFORE CREATION).
- **FINAL STATES:** COMPLETED, CANCELLED.
- **STATES:**
 - **SCHEDULED:** SLOT IS BOOKED BUT PAYMENT/CONSULTATION PENDING.
 - **CONFIRMED:** PAYMENT SUCCESSFUL, WAITING FOR TIME SLOT.
 - **IN-PROGRESS:** VIDEO CALL IS ACTIVE.
 - **COMPLETED:** CALL FINISHED, PRESCRIPTION GENERATED.
 - **CANCELLED:** USER OR DOCTOR CANCELLED THE SLOT.

2. TRANSITIONS & EVENTS:

- NULL -> SCHEDULED [EVENT: PATIENT SELECTS SLOT].
- SCHEDULED -> CONFIRMED [EVENT: PAYMENT SUCCESSFUL].
- SCHEDULED -> CANCELLED [EVENT: PAYMENT FAILED / TIMEOUT].
- CONFIRMED -> IN-PROGRESS [EVENT: MEETING TIME STARTS & JOIN LINK CLICKED].
- CONFIRMED -> CANCELLED [EVENT: PATIENT CANCELS > 24HRS PRIOR].
- IN-PROGRESS -> COMPLETED [EVENT: DOCTOR ENDS CALL].

3. STATE TRANSITION DIAGRAM (STD):

